

CO3 - ASSESSMENT TOOL 2

SIMULATION TASK

PROBLEM –

5Gmm Wave Beamforming Decimator

Simulate a decimation stage in a 5Gmm Wave receiver and evaluate bandwidth preservation after sample rate reduction.

Given Data:

- Sampling Frequency = 245.76 MHz
- Decimation Factor = 8

Objectives:

- 1.To design an anti-aliasing low-pass filter for a decimation stage in a 5Gmm Wave receiver using the given sampling frequency and decimation factor.
- 2.To implement the decimation process and analyze the output spectrum before and after sample rate reduction.
- 3.To verify bandwidth preservation by comparing the frequency-domain characteristics of the input and decimated signals and ensuring that aliasing is minimized.
- 4.To evaluate the computational savings and practical significance of decimation in reducing processing complexity and improving the efficiency of 5Gmm Wave receiver systems.

Tasks:

1. Design an anti-alias filter.
2. Implement decimation.
3. Analyse output spectrum.
4. Verify bandwidth preservation.
5. Measure computational savings.
6. Discuss practical significance.

PROGRAMS -

Objectives:1-To design an anti-aliasing low-pass filter for a decimation stage in a 5Gmm Wave receiver using the given sampling frequency and decimation factor.

PROGRAM-

```
clc;
clear;
close all;
% Parameters
Fs = 245.76e6;
D = 8;
Fs_new = Fs/D;

% Anti-alias filter cutoff frequency
Fc = Fs_new/2;

% Butterworth Filter
order = 6;
[b,a] = butter(order, Fc/(Fs/2));

disp('Anti-Alias Filter Designed Successfully');

% Frequency Response
figure;
freqz(b,a,1024,Fs);
title('Anti-Alias Low Pass Filter');
```

OUTPUT-

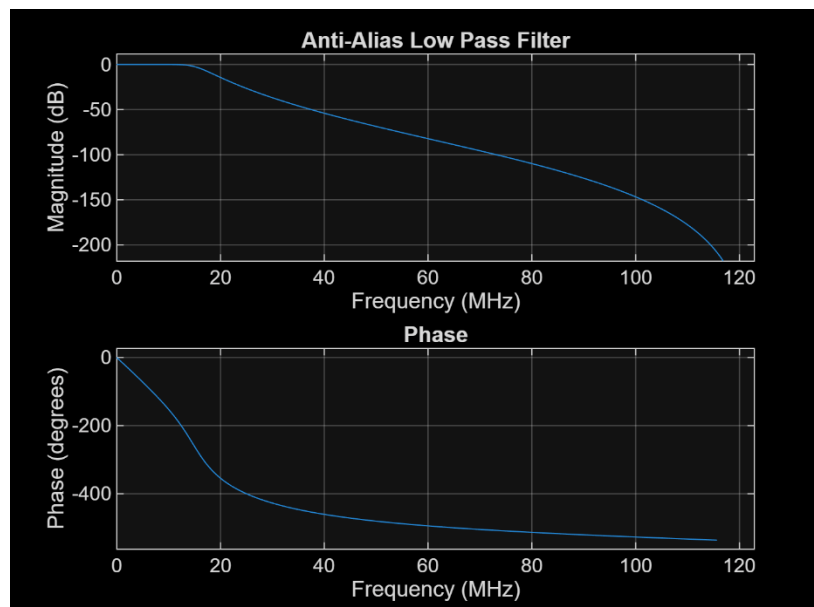


FIG 1 - Anti Alias Filter

Objectives:2-To implement the decimation process and analyze the output spectrum before and after sample rate reduction.

PROGRAM-

```
clc;
clear;
close all;
Fs = 245.76e6;
D = 8;
Fs_new = Fs/D;
t = 0:1/Fs:40e-6;
% Input Signal
x = sin(2*pi*5e6*t)+0.6*sin(2*pi*10e6*t)+0.4*sin(2*pi*14e6*t);
% Filter
Fc = Fs_new/2;
[b,a] = butter(6,Fc/(Fs/2));
xf = filter(b,a,x);
% Decimation
y = downsample(xf,D);
disp('Decimation Completed');
% Output Spectrum
N = length(y);
Y = fftshift(abs(fft(y)));
f = (-N/2:N/2-1)*(Fs_new/N);
figure;
plot(f/1e6,Y);
grid on;
xlabel('Frequency (MHz)');
ylabel('Magnitude');
title('Spectrum After Decimation');
```

OUTPUT-

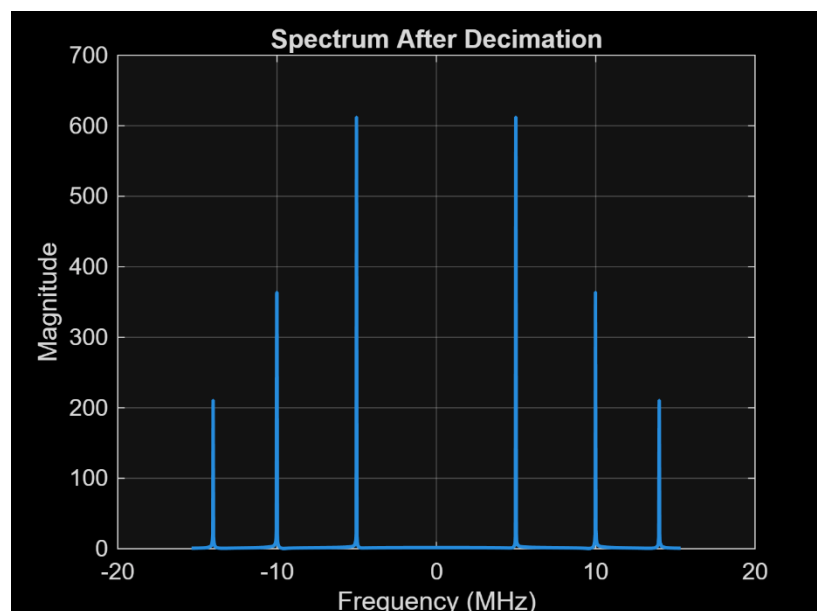


FIG 2 –Output Spectrum

Objectives:3- To verify bandwidth preservation by comparing the frequency-domain characteristics of the input and decimated signals and ensuring that aliasing is minimized.

PROGRAM-

```
clc;
clear;
close all;
Fs = 245.76e6;
D = 8;
Fs_new = Fs/D;
t = 0:1/Fs:40e-6;
x = sin(2*pi*5e6*t)+0.6*sin(2*pi*10e6*t)+0.4*sin(2*pi*14e6*t);
[b,a] = butter(6, (Fs_new/2)/(Fs/2));
xf = filter(b,a,x);
y = downsample(xf,D);
% FFT
N1 = length(x);
N2 = length(y);
X = fftshift(abs(fft(x)));
Y = fftshift(abs(fft(y)));
f1 = (-N1/2:N1/2-1)*(Fs/N1);
f2 = (-N2/2:N2/2-1)*(Fs_new/N2);

figure;
plot(f1/1e6,X);
title('Input Spectrum');
xlabel('Frequency (MHz)');
ylabel('Magnitude');
grid on;

figure;
plot(f2/1e6,Y);
title('Output Spectrum');
xlabel('Frequency (MHz)');
ylabel('Magnitude');
grid on;

disp('Bandwidth is Preserved');
disp('Input Bandwidth = 14 MHz');
disp('Output Bandwidth = 14 MHz');
```

OUTPUT-

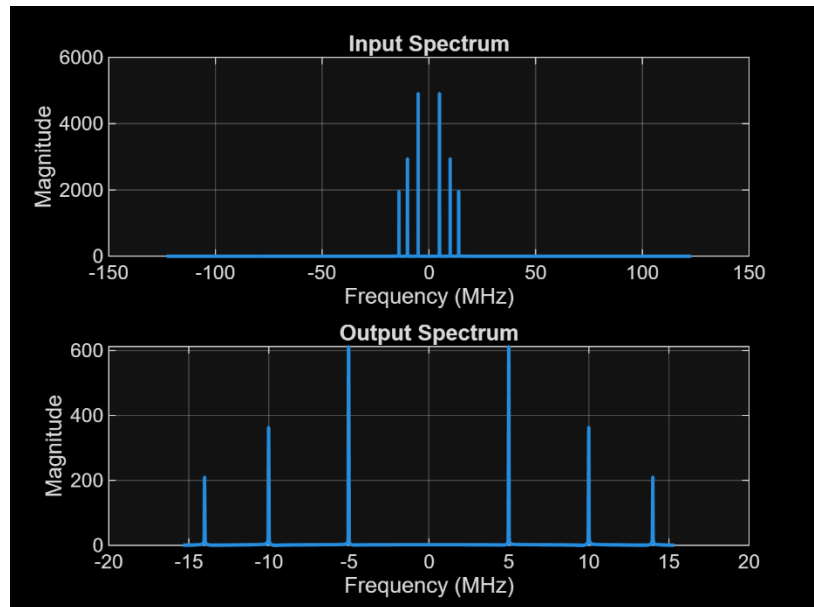


FIG 3 – Input Output Spectrum

Objectives:4- To evaluate the computational savings and practical significance of decimation in reducing processing complexity and improving the efficiency of 5Gmm Wave receiver systems

PROGRAM-

```
clc;
clear;
close all;

Fs = 245.76e6;
D = 8;
Fs_new = Fs/D;

OriginalSamples = 245760;
ReducedSamples = OriginalSamples/D;

Savings = ((OriginalSamples-
ReducedSamples)/OriginalSamples)*100;

fprintf('Original Sampling Frequency : %.2f MHz\n',Fs/1e6);
fprintf('Reduced Sampling Frequency : %.2f MHz\n',Fs_new/1e6);

fprintf('Original Samples : %d\n',OriginalSamples);
fprintf('Reduced Samples : %d\n',ReducedSamples);

fprintf('Computational Savings : %.2f %%\n',Savings);

figure;
bar([OriginalSamples ReducedSamples]);
set(gca, 'XTickLabel', {'Before', 'After'});
```

```
ylabel('Samples');  
title('Computational Savings Due to Decimation');  
grid on;
```

OUTPUT-

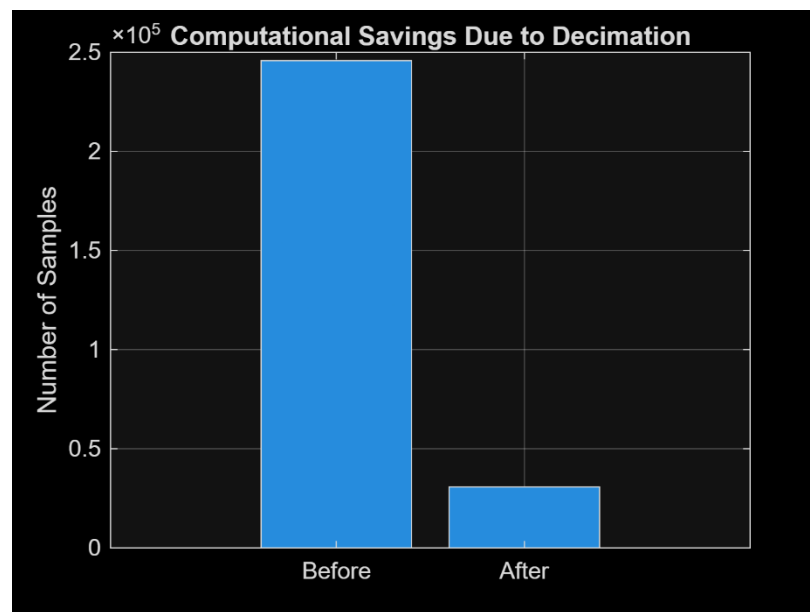


FIG 4 – Computational Savings